

Problema 2: 50%

Para las siguientes CFGs realice lo solicitado, mostrando todo el procedimiento.

~~1.~~

$$S \rightarrow 0A0 \mid 1B1 \mid BB$$

$$A \rightarrow C$$

$$B \rightarrow S \mid A$$

$$C \rightarrow S \mid \epsilon$$

~~2.~~

$$S \rightarrow aAa \mid bBb \mid \epsilon$$

$$A \rightarrow C \mid a$$

$$B \rightarrow C \mid b$$

$$C \rightarrow CDE \mid \epsilon$$

$$D \rightarrow A \mid B \mid ab$$

3.

$$S \rightarrow ASA \mid aB$$

$$A \rightarrow B \mid S$$

$$B \rightarrow b \mid \epsilon$$

- a) Elimine las producciones- ϵ .
- b) Elimine cualquier producción unaria de la gramática resultante.
- c) Elimine cualquier símbolo inútil de la gramática resultante.
 - a. Remueva símbolos que no producen.
 - b. Remueva símbolos no alcanzables.
- d) Coloque la CFG resultante en la Forma Normal de Chomsky (CNF).

1.

$$S \rightarrow 0A0 \mid 1B1 \mid BB$$

$$A \rightarrow C$$

$$B \rightarrow S \mid A$$

$$C \rightarrow S \mid \epsilon$$

$$\text{Nulable} = \{S, A, B, C\}$$

$$S_0 \rightarrow S \mid \epsilon$$

$$S \rightarrow 0A0 \mid 00 \mid 1B1 \mid 11 \mid B \mid BB$$

$$A \rightarrow C$$

$$B \rightarrow S \mid A$$

$$C \rightarrow S$$

Unitarias: $S_0 \rightarrow S$, $S \rightarrow B$, $A \rightarrow C$, $B \rightarrow S$, $B \rightarrow A$, $C \rightarrow S$.

eliminamos unitarias:

$$S_0 \rightarrow \epsilon \mid 0A0 \mid 00 \mid 1B1 \mid 11 \mid BB$$

$$S \rightarrow 0A0 \mid 00 \mid 1B1 \mid 11 \mid BB$$

$$A \rightarrow 0A0 \mid 00 \mid 1B1 \mid 11 \mid BB$$

$$B \rightarrow 0A0 \mid 00 \mid 1B1 \mid 11 \mid BB$$

$$C \rightarrow 0A0 \mid 00 \mid 1B1 \mid 11 \mid BB$$

productivos: $\{S_0, S, A, B, C\}$

no alcanzables: $\{S, C\}$

$$S_0 \rightarrow 0A0 \mid 00 \mid 1B1 \mid 11 \mid BB \mid \epsilon$$

$$A \rightarrow 0A0 \mid 00 \mid 1B1 \mid 11 \mid BB$$

$$B \rightarrow 0A0 \mid 00 \mid 1B1 \mid 11 \mid BB$$

$$X_0 \rightarrow 0$$

$$0A0 \rightarrow X_0 A X_0$$

$$00 \rightarrow X_0 X_0$$

$$Y_0 \rightarrow A X_0$$

$$0A0 \rightarrow X_0 Y_0$$

$$X_1 \rightarrow 1$$

$$1B1 \rightarrow X_1 B X_1$$

$$11 \rightarrow X_1 X_1$$

$$Y_1 \rightarrow B X_1$$

$$1B1 \rightarrow X_1 Y_1$$

CFG

$$S \rightarrow X_0 Y_0 \mid X_0 X_0 \mid X_1 Y_1 \mid X_1 X_1 \mid BB \mid \epsilon$$

$$A \rightarrow X_0 Y_0 \mid X_0 X_0 \mid X_1 Y_1 \mid X_1 X_1 \mid BB$$

$$B \rightarrow X_0 Y_0 \mid X_0 X_0 \mid X_1 Y_1 \mid X_1 X_1 \mid BB$$

$$Y_0 \rightarrow A X_0 \quad X_1 \rightarrow 1$$

$$Y_1 \rightarrow B X_1$$

$$X_0 \rightarrow 0$$

2.

$$S \rightarrow aAa \mid bBb \mid \epsilon$$

$$A \rightarrow C \mid a$$

$$B \rightarrow C \mid b$$

$$C \rightarrow CDE \mid \epsilon$$

$$D \rightarrow A \mid B \mid ab$$

$$\text{Nulable} = \{S, A, B, C, D\}$$

$$S_0 \rightarrow S \mid \epsilon$$

$$S \rightarrow aAa \mid aa \mid bBb \mid bb$$

$$A \rightarrow C \mid a$$

$$B \rightarrow C \mid b$$

$$C \rightarrow CDE \mid DE \mid CE \mid E$$

$$D \rightarrow A \mid B \mid ab$$

Unitarias: $S_0 \rightarrow S$, $A \rightarrow C$, $B \rightarrow C$, $C \rightarrow E$, $D \rightarrow A$, $D \rightarrow B$.

eliminamos unitarias:

$$S_0 \rightarrow aAa \mid aa \mid bBb \mid bb \mid \epsilon$$

$$S \rightarrow aAa \mid aa \mid bBb \mid bb$$

$$A \rightarrow aCDE \mid DE \mid CE \mid E$$

$$B \rightarrow bCDE \mid DE \mid CE \mid E$$

$$C \rightarrow CDE \mid DE \mid CE$$

$$D \rightarrow CDE \mid DE \mid CE \mid a \mid b \mid ab$$

productivos: $\{S_0, S, A, B, D\}$

no alcanzables: $\{E, C\}$

$$S_0 \rightarrow aAa \mid aa \mid bBb \mid bb \mid \epsilon$$

$$S \rightarrow aAa \mid aa \mid bBb \mid bb$$

$$A \rightarrow a, \quad B \rightarrow b$$

$$D \rightarrow ab \mid a \mid b$$

$$S_0 \rightarrow aAa \mid aa \mid bBb \mid bb \mid \epsilon$$

$$A \rightarrow a$$

$$B \rightarrow b$$

$$\text{CFG: } S_0 \rightarrow AX \mid AA \mid BY \mid BB \mid \epsilon$$

$$X \rightarrow AA$$

$$Y \rightarrow BB$$

$$A \rightarrow a$$

$$B \rightarrow b$$

3.

$$S \rightarrow ASA \mid aB$$

$$A \rightarrow B \mid S$$

$$B \rightarrow b \mid \epsilon$$

$$\text{Nullable} = \{A, B\}$$

$$S \rightarrow A\epsilon A \mid SA \mid AS \mid S \mid aB \mid a$$

$$A \rightarrow B \mid S$$

$$B \rightarrow b$$

Unitarias: $S \rightarrow S$, $A \rightarrow B$, $A \rightarrow S$

eliminamos unitarias:

$$S \rightarrow A\epsilon A \mid SA \mid AS \mid S \mid aB \mid a$$

$$A \rightarrow B \mid S$$

$$B \rightarrow b$$

productivos: $\{S, A, B\}$
no alcanzables: $\{\emptyset\}$

CFG

$$S \rightarrow AX_{SA} \mid SA \mid AS \mid X_a B \mid a$$

$$A \rightarrow AX_{SA} \mid SA \mid AS \mid X_a B \mid a \mid b$$

$$B \rightarrow b$$

$$X_{SA} \rightarrow SA$$

$$X_a \rightarrow a$$